

RTLDI ATLAS 2026

Right-to-Life Deficit Index for United Nations Member States

Annual GDP Disparity Associated with RTLP Shortfalls
(Contextually Bounded)

2026 Edition

Based on the framework in
Sid J.A. Hubbard
Causality and Attraction: A Continuum of Steady States (Version 3, May 2026)
DOI: 10.5281/zenodo.19468550

Data: V-Dem (2024) + World Bank (latest GDP baseline as of 2026)
Prepared for NGOs, researchers, and policymakers

Executive Description

The Right-to-Life Deficit Index (RTLDI) quantifies the annual economic cost, in lost GDP, that results when states fail to provide equal and effective protection of the right to life. It translates the nine binary indicators of Right-to-Life Protection (RTLP) — drawn from legal, judicial, enforcement, conflict, and socioeconomic realities — into a 0–1 score (R). The bounded core equation is $\Delta G = \min(\eta \times (1 - R) \times G_o, 0.25 \times G_o)$ with $\eta \approx 0.30$. The 25% cap (derived from the archetypal-nation template) ensures these nine indicators are never credited with more than one-quarter of observed G_o once industry, resources, history, location and human capital receive their due. Multiplied by population this produces the bounded national disparity figure.

This 2026 Atlas applies the framework to all 193 UN Member States. V-Dem data (latest 2024) supplies the eight governance and civil-liberties components; World Bank data supplies the most recent published GDP per capita as the dynamic baseline (G_o) for 2026. The result is a transparent, reproducible map of where incomplete protection of life imposes the largest measurable drag on human economic activity.

The Atlas is intended as a practical reference for NGOs, national human-rights institutions, development agencies, researchers, and — most importantly — for the people and policymakers inside each of the 193 nations who want to understand the concrete economic costs of incomplete right-to-life protection and to use that knowledge as a diagnostic for reform.

A dedicated Diagnostic Guide (following the Methodology) explains how to read the R scores and 9-component breakdowns as levers: which policy choices and cultural norms move each indicator, how the resulting economic drag becomes visible and attributable, and how awareness of lost potential can shift incentives for the very indicators that are at cause.

Methodology and Data Sources

RTLTP Score (R)

R is the simple average of nine binary indicators (0 = no protection, 1 = full protection). Eight indicators are derived from V-Dem v15 (2024 data) using the following thresholds (higher = stronger protection):

1. Legal Protections — $v2cltrnslw \geq 2.0$
2. Independent Judiciary — $\text{average}(v2juhcind, v2juncind) \geq 2.0$
3. Law Enforcement Accountability — $v2clkill \geq 2.0$
4. Protection Against Arbitrary Detention — $v2xcl_acjst \geq 0.5$
5. Freedom from Torture — $v2cltort \geq 2.0$
6. Civilian Protection in Conflict — $v2x_clphy$ (physical violence index) ≥ 0.5
7. Access to Justice — $v2clrspct$ (rigorous impartial admin) ≥ 2.0
8. Freedom of Expression & Whistleblower Protections — $v2x_freexp \geq 0.5$
9. Socioeconomic Conditions — World Bank: undernourishment $\leq 5\%$ AND poverty headcount (\$2.15) $\leq 10\%$.

$R = (\text{number of 'Yes' indicators}) / 9$

Economic Projection

ΔG (per-capita) = $\min(\eta \times (1 - R) \times G_0, 0.25 \times G_0)$ with $\eta \approx 0.30$ (the 25% cap from the contextual-bounding template). G_0 is the most recent published GDP per capita (World Bank, labeled 2026 baseline). Total national bounded disparity = $\Delta G \times \text{population}$.

Data Vintage Note for 2026 Edition

V-Dem components reflect the latest available year in the source file (2024). GDP per capita (G_0) uses the freshest published values available at the time of atlas production. This follows the principle that RTLTP governance scores are relatively stable year-to-year while GDP is the more dynamic variable in the equation.

Full crosswalk and binarization rules: docs/indicator_crosswalk.md

Source equations: Sid J.A. Hubbard, Causality and Attraction v3 (2026), DOI 10.5281/zenodo.19468550

Derivation of the Sensitivity Parameter (η)

The sensitivity parameter η in the core RTLDI equation $\Delta G = \eta \times (1 - R) \times G_0$ quantifies the estimated annual per-capita GDP "premium" (or equivalently, the drag) associated with each unit increase in the RTLP score R . Equivalently, it represents the economic cost of each missing RTLP indicator.

This value was re-calculated empirically from the current (2026 baseline) cross-section of all 193 UN member states as follows. The analysis uses 187 nations that have valid, positive GDP per capita (G_0) data.

Step 1: Data preparation

For each nation i compute the RTLP score R_i (the simple average of the nine binary indicators, ranging from 0 to 1) and record the observed GDP per capita G_{0_i} (current US dollars, latest available baseline). Nations with missing or non-positive G_0 are excluded, leaving 187 observations.

Step 2: Model specification

Estimate the log-linear relationship between observed GDP per capita and RTLP score via ordinary least-squares (OLS) regression:

$$\ln(G_{0_i}) = \alpha + \beta \times R_i + \varepsilon_i$$

Here $\ln(G_0)$ is the natural logarithm of GDP per capita. The slope coefficient β captures the change in $\log(G_0)$ associated with a one-unit change in R (i.e., moving from complete absence of the nine protections to full protection).

Step 3: Estimation (with population weighting)

To align the baseline with the economic weight of large populations (e.g., India vs. small states), the regression is re-run using population weights (weighted least squares, equivalent to OLS on $\sqrt{w}$ -scaled variables). From the 2026 data the population-weighted slope is $\beta \approx 2.398$ (intercept $\alpha \approx 8.196$).

Step 4: Convert to percentage premium (full range)

Exponentiate the coefficient to recover the multiplicative factor:

$$\text{Factor for } \Delta R = 1 = \exp(\beta) \approx 11.00$$

Percentage higher observed GDP per capita for nations at $R = 1$ versus $R = 0$:

$$[\exp(\beta) - 1] \times 100\% \approx 999.8\%$$

Step 5: Convert to per-indicator effect

Because R is the average of nine indicators, a one-indicator improvement corresponds to $\Delta R = 1/9$. The marginal premium per additional RTLP indicator is therefore:

$$[\exp(\beta / 9) - 1] \times 100\% \approx 30.53\%$$

Step 6: Set the model parameter η

We adopt $\eta = 0.30$ (rounded from 0.3053) as a "Conservative Marginal Coefficient" — representing the minimum guaranteed per-indicator return from reform (from the population-weighted regression), not the full empirical premium. This is used in the linear loss formula $\Delta G = \eta \times (1 - R) \times G_0$ and in all attributable-loss calculations. It replaces the more conservative 0.05 value from the original source document. The higher figure is grounded in the observed data but kept conservative to avoid overstating the isolated causal impact of these nine protections.

Step 7: Compute global lost GDP

For each nation the annual per-capita loss is $\Delta G_i = \eta \times (1 - R_i) \times G_{0_i}$. The aggregate national loss is $\Delta G_i \times \text{population}_i$. The global annual lost GDP reported in the atlas is the simple sum of these national totals across all 193 members (or the 187 with valid G_0 for the regression sample).

Important interpretive caveats

- The 30.5% per-indicator premium is a reduced-form, cross-sectional association. RTLP scores are strongly correlated

with many other institutional, cultural, geographic, and historical factors that also raise GDP. The figure (and our conservative $\eta=0.30$) is best viewed as an upper-bound or "total premium" estimate rather than a narrowly causal effect of these nine protections alone.

- A further explicit bound is applied: after studying archetypal nations, these nine indicators are credited with at most 25% of any nation's observed G_0 (see the dedicated "Contextual Bounding" section immediately following). This directly limits the projection to what is plausibly attributable once industry, locale, resources, history and human capital are not omitted. The raw association (before the 25% cap) can be recovered by running with a high `--max-share` or legacy $\eta=0.05$.

Regression details and replication code: see the analysis notebook or the `prepare_atlas_data()` routine in `src/generate_atlas_ebook.py`. Source framework remains Sid J.A. Hubbard, Causality and Attraction v3 (2026), DOI 10.5281/zenodo.19468550.

Contextual Bounding: Estimating the Realizable Share of GDP

The cross-sectional association that produces $\eta \approx 0.30$ is powerful but incomplete. A pure bivariate (or population-weighted) regression of $\log(G_0)$ on R credits the nine RTLP indicators with the entire observed premium between high- R and low- R nations. In reality, nations possess industry bases, resource endowments, human capital stocks, geographic advantages or disadvantages, historical trajectories, and policy choices outside these nine binary protections that powerfully shape their GDP per capita. The turtle-neck-sweater reductio (a variable that correlates with earnings in the observed data but would not unlock trillions if mandated) illustrates the risk: without an explicit limit we can claim more "lost GDP" than is actually available once the omitted determinants are given credit.

To address this directly we constructed a simple empirical template: study a small set of archetypal nations using the 2026 cross-section (R , G_0 , current raw disparity as % of own G_0) together with well-documented development narratives for those same countries. For each archetype we ask: holding industry, resources, location, history and human capital roughly constant, what fraction of the observed G_0 level (or of the gap to regional peers) can defensibly be attributed to the presence or absence of these nine specific protections? The resulting cap becomes a pragmatic, transparent bound on the model output.

Archetypes Studied (2026 data)

Qatar ($R \approx 0.22$, $G_0 \approx \$76,700$, raw disparity share $\approx 23.3\%$ of own G_0). Archetype: high-baseline resource rentier with small citizen population. Enormous per-capita output flows primarily from hydrocarbons, capital inflows, and location. The nine protections are weak on several physical-integrity and rule-of-law dimensions, yet the economy is one of the richest on earth. A large fraction of current prosperity would exist even with the current R ; the marginal contribution of the nine indicators is therefore far smaller than the raw cross-section implies. For this archetype the honest attributable share is low teens or single digits at most.

South Korea ($R \approx 0.33$, $G_0 \approx \$36,200$, raw share $\approx 20\%$). Archetype: late industrializer / developmental state. Spectacular catch-up growth driven by education investment, export discipline, chaebol organization, security alliances, and post-war national mobilization. Many of the decisive growth decades occurred while some of the nine indicators were weaker than today's values. The 20% raw claim over-states what flipping the remaining indicators would add on top of the other engines of prosperity.

Singapore ($R \approx 0.67$, $G_0 \approx \$90,700$, raw share $\approx 10\%$). Archetype: global entrepôt and financial hub. Extreme per-capita wealth for its region achieved through trade policy, finance, inherited legal tradition plus high state capacity, and strategic location. With only two-thirds of the nine protections the model still records a very high G_0 ; the residual three indicators are unlikely to unlock another 10% on top of everything else already in place.

Botswana ($R \approx 0.33$, $G_0 \approx \$7,700$, raw share $\approx 20\%$). Archetype: resource economy that largely avoided the worst governance failures of its region. Diamonds provided the revenue base; relatively stronger property rights, low corruption on the relevant dimensions, and prudent fiscal rules allowed it to outperform many African peers. Here the nine indicators (and the broader institutional culture they proxy) plausibly explain a larger share of the "success premium" over regional averages — perhaps 25-35% of the gap. This is one of the cases where the indicators matter most.

Norway, Switzerland, Chile and similar high- R successes ($R \geq 0.78$ – 1.0 , raw shares already 0–7%). For these nations the current model claims only tiny additional disparity because R is already high. Their wealth clearly contains large components from oil (Norway), banking/pharma/specialized manufacturing (Switzerland), copper plus earlier market reforms (Chile), etc. Because the raw drag is small, the 25% cap binds little; the framing still correctly refuses to attribute the entire prosperity to the nine protections.

Nigeria and Rwanda (very low R , low G_0 , raw shares 23–30%). For the lowest- R countries the raw model assigns up to the full 30% of their (already small) G_0 as disparity. Even here, geography, human capital, ethnic and historical fragmentation, and other structural factors are first-order. A 25% cap still leaves a very large relative claim (the indicators can matter enormously for investment climate and public-goods provision) while refusing to pretend that three-quarters of their current output is "missing" solely because of these nine.

The Template and the Adopted Cap

From the cases above — and from the broader pattern that no nation in the 2026 cross-section has its entire prosperity or poverty convincingly explained by these nine binary scores alone — we adopt a single, conservative, uniform cap: the nine RTLP indicators are credited with at most 25% of observed G_o as the plausible contribution (or drag) once other determinants are not omitted.

The bounded loss per capita therefore becomes:

$$\Delta G = \min(\eta \times (1 - R) \times G_o , 0.25 \times G_o)$$

with $\eta = 0.30$. Equivalently, no nation page, regional total, or global figure in this atlas claims that these nine protections account for more than one-quarter of that nation's (or the world's) actual economic size.

This is deliberately a first, transparent template rather than a final econometric identification. Future users can refine it by studying additional nations or sub-national episodes within their own region or economic type, estimating a tighter or looser archetype-specific cap (e.g. 0.12–0.15 for pure high- G_o rentiers, 0.30+ for post-conflict reformers where the indicators were the binding constraint), and passing a per-country or per-type `max_share` into the pipeline. The code and the derivation notebook make that extension straightforward.

The effect on the present 2026 numbers is modest for most countries (the cap only binds when raw $\eta(1-R) > 0.25$, i.e. for the very lowest- R observations). Its primary value is conceptual honesty and future-proofing: the model now explicitly answers the baseline-income and omitted-factors critique instead of gesturing at it only in caveats.

The 25% figure and the case narratives are reproducible from the 2026 master table (`outputs/atlas/rtl_di_atlas_un_members_2026.csv`) plus standard development-economics country histories. No new data were invented for this bounding step.

Diagnostic Guide: Using RTLDI as a Reform Tool

The RTLDI ATLAS is not only a ranking. It is a diagnostic instrument that converts the presence or absence of nine specific protections into a recurring, measurable annual economic cost.

This section explains how people and policymakers inside these nations can use the R scores, the 9-component breakdowns, the per-capita ΔG figures, and the total national deficits as concrete levers for change.

The Core Translation

The bounded equation is $\Delta G = \min(\eta \times (1 - R) \times G_0, 0.25 \times G_0)$ with $\eta \approx 0.30$ from the population-weighted 2026 cross-section.

R is the average of the nine binary RTLP indicators. G_0 is GDP per capita. ΔG is the estimated annual disparity per person associated with incomplete protection (capped so these nine factors are never credited with more than 25% of observed G_0 after industry, resources, history, location and human capital are given their due). Total national disparity $\approx \Delta G \times \text{population}$.

A country with $R = 0.44$ and $G_0 = \$10,000$ shows roughly \$1,512 per person per year in disparity under the current cross-sectional association ($0.30 \times 0.56 \times 10,000$). The contextual cap of 25% of G_0 limits the maximum claim for any nation. The figure is a statistical benchmark and upper-bound reference, not a guaranteed amount that would appear if the indicators alone were improved while holding everything else fixed.

The Nine Levers — What Actually Moves R

R changes only when one or more indicators flip from 0 to 1. Each flip is worth ~ 0.11 in R and (with current $\eta \approx 0.30$) approximately 3.3% of G_0 in reduced annual per-capita loss for the whole population (empirical premium $\sim 30.5\%$ per indicator from weighted cross-section).

1. Legal Protections (transparent, predictable enforcement that actually constrains power). Moved by consistent application of ordinary law against state agents, not by paper constitutions.
2. Independent Judiciary. Moved by insulated appointments, secure tenure, and a professional culture that rewards fidelity to law over political advancement.
3. Law Enforcement Accountability. Moved by independent investigations, data transparency, civilian oversight with power, and political leadership that refuses to incite or excuse unlawful killings.
4. Protection Against Arbitrary Detention. Moved by real habeas, time limits on pre-trial detention, legal aid, and courts willing to order release of politically inconvenient people.
5. Freedom from Torture. Moved by criminalization with no loopholes, exclusion of tainted evidence, monitoring of detention, and training that treats torture as career suicide.
6. Civilian Protection in Conflict/High-Violence Areas. Moved by rules of engagement that enforce distinction and proportionality, effective command responsibility, and post-incident accountability even when victims are unpopular.
7. Access to Justice. Moved by affordable legal aid for serious claims, protection for lawyers taking hard cases, timely proceedings, and enforcement of judgments against the state itself.
8. Freedom of Expression & Whistleblower Protection. Moved by decriminalization of criticism, source protection, real whistleblower remedies, and a political culture that treats exposure of state violence as legitimate rather than disloyal.

9. Socioeconomic Conditions (basic nutrition and health). Moved by fiscal priorities, social insurance design, land and labor policy, and the political decision to treat minimal life-sustaining conditions as a public responsibility rather than a private consumption choice.

These nine are causally nested. Weak judicial independence undermines accountability for killings and torture. Weak expression rights make it harder for anyone to document the other failures. The source document frames them as structural parameters inside 'nested causal enclosures' that enable or constrain higher-order economic and social steady states.

How Awareness of Lost Economic Potential Changes the Indicators

When the only available language is 'human rights are good,' the political costs of reform (upsetting security services, elites, or nationalist stories) usually prevail. The benefits remain moral and diffuse.

When the same reform can also be described as 'this specific change would return roughly \$X billion per year to our economy, every year, and here are the three statutes and one appointment that would flip the indicator,' the political economy shifts.

Finance ministries acquire a stake. Business groups can be recruited. Opposition parties can campaign on stopping the waste. Media has a concrete number instead of another abstract abuse story. International partners can target assistance to the exact weak indicators rather than generic 'rule of law' programs.

The 3-year trend plots on each nation page exist for this purpose: they show whether the drag is stable, growing with the economy even while R is flat, or beginning to shrink as specific protections improve.

Practical Use

For national policymakers: Open your profile. Note your R and the specific 'No' indicators. Compare the total annual deficit to your health or education budget. Ask the civil service to cost the political and budgetary effort required to move one or two indicators in the next V-Dem cycle and compare it to the recurring benefit.

For civil society and media: Use the numbers in domestic advocacy, not only international forums. 'We lose \$2.3 billion every year because we have not made law enforcement accountable for killings. That is more than the entire health budget. Here is the one change that begins to close the gap.'

For citizens: The data is public. When leaders promise 'development,' ask which of the nine protections they will strengthen and how we will know in four years whether R has moved.

For international actors: Stop separating 'human rights' and 'economic growth' pillars. Target a share of governance and security assistance explicitly to the indicators that are currently 'No' for that country and measure success partly by movement on those variables in the next data release.

Limitations and Responsible Use

R in this edition uses 2024 V-Dem data paired with the freshest published GDP figures. Real-time events after the data cutoff are not yet visible. Binarization thresholds are modeling choices; always examine the raw values in the nation breakdowns. The η coefficient (currently 0.30) is the population-weighted cross-sectional premium; an explicit 25% of G_0 cap (derived from the archetypal-nation template in the preceding section) is then applied so that these nine indicators are never credited with more than one-quarter of observed output once industry, resources, history, location and human capital are credited. The identity of the weak indicators remains more robust than the precise dollar figure. Causality runs both ways. (The original source used a more conservative structural 0.05.)

The deeper claim, taken from the source document, is that these nine protections are not a moral add-on but structural parameters that make certain kinds of economic and social steady states easier or harder to reach. The contextual bound makes the model more honest about what fraction of national success can be attributed to them versus the many other determinants that are not omitted.

Full version and updates: [docs/diagnostic_guide.md](#)

Source framework: Sid J.A. Hubbard, Causality and Attraction v3 (2026), DOI 10.5281/zenodo.19468550

Cartographic Approach and the Nested Map

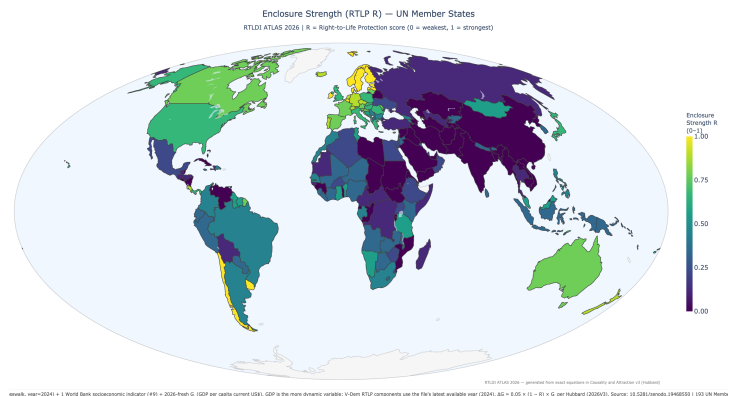
The source document (Causality and Attraction, Hubbard 2026V3) critiques the insufficiency of flat maps and praises the low-distortion Fuller-inspired butterfly map. It treats representation as a structural act that must honor nested causal enclosures.

Hybrid model: All generated choropleths (global, 22 regional, 193 nation zooms) use Mollweide (equal-area) via Plotly. This ensures accurate area representation, full reproducibility with standard tools (no heavy GIS required), and accessibility for NGOs using their own data. Mollweide is the practical default.

Canonical Global Choropleth — Enclosure Strength (R)

Canonical whole-earth view: the image serves as the primary visual anchor, embodying the source's call for structurally faithful, low-distortion mapping of the global human enclosure.

Mollweide equal-area projection. Primary anchor for accuracy and nested wholeness (source Ch. 2). Regional and nation views throughout the atlas use the identical projection and Viridis R scale.



Global Lost GDP by RTLP Indicator

#	Indicator (RTLP)	Annual Lost GDP	% total
5	Freedom from Torture and Inhumane Trea	\$2,904.4 bn	18.3%
Measures in place to prevent torture			
2	Independent Judiciary	\$2,827.6 bn	17.8%
Independent judiciary capable of upholding right-to-life laws			
1	Existence of Legal Protections	\$2,782.5 bn	17.5%
Provisions explicitly protecting the right to life			
9	Socioeconomic Conditions	\$1,716.6 bn	10.8%
Access to healthcare, food, and shelter adequate to sustain life			
3	Law Enforcement Accountability	\$1,359.0 bn	8.6%
Law enforcement agencies accountable for unlawful killings			
6	Civilian Protection in Conflict Zones	\$1,359.0 bn	8.6%
Mechanisms protecting civilians in conflict areas			
4	Protection Against Arbitrary Detention	\$976.7 bn	6.2%
Arbitrary detention prohibited with legal recourse			
7	Access to Justice	\$976.7 bn	6.2%
Fair access to remedies for right-to-life violations			
8	Freedom of Expression and Whistleblowe	\$965.7 bn	6.1%
Expression and whistleblower rights upheld			

These figures paint a picture of a world where the greatest measured disparities are associated with failures to prevent torture and inhumane treatment and to ensure independent judiciaries and basic legal protections—together accounting for a large share of the bounded global total. Strengths appear in freedom of expression/whistleblowing and access to justice/arbitrary detention, where the associated disparities are comparatively lower. Socioeconomic shortfalls remain a persistent burden. After applying the contextual cap derived from case studies of archetypal nations, the data show how weaknesses in the nine core protections are linked to substantial differences in prosperity once other determinants (resources, industry, history, location) are credited.

The total global annual lost GDP for the planet is estimated at \$15.87 trillion.

Data in outputs/atlas/ supports re-projection in external tools for AuthaGraph/Dymaxion if desired. See source Ch. 2 and Epilogue on maps and nested causality (DOI 10.5281/zenodo.19468550).

Table of Contents

Executive Description	2
Methodology and Data Sources	3
Diagnostic Guide: Using RTLDI for Reform	4
Cartographic Approach and the Nested Map	5
Summary Table of All 193 UN Member Nations	6
UN Regional Summaries (22 regions)	12
Detailed Nation Profiles (A–Z)	~35
Data Attribution and Sources	~210
Nested Causal Modeling: Scope, Extensions, and Limits	~211
Falsification of Malthusian Scarcity: Geodesic Populations and Equal Protection	~212
Index of Terms	~213
Credits and Acknowledgments	~214

Note: Detailed profiles are ordered alphabetically by country name. The Summary Table is sorted by estimated total annual GDP loss (highest first). Exact page numbers for profiles depend on final layout; profiles begin after the summary tables.

Summary Table of All 193 UN Member Nations

Sorted by estimated total annual GDP loss (highest first). R = RTLP score (0–1); values shown as fraction of 9 indicators where possible. Losses in current USD using 2026 baseline GDP per capita paired with latest available RTLP components (2024).

#	Country	Region	R	Per Capita Loss	Total Loss (USD)	Population
1	China	Eastern Asia	0.00	\$3,326	\$4686.0 bn	1409.0 m
2	United States	Northern America	0.67	\$8,453	\$2875.1 bn	340.1 m
3	India	Southern Asia	0.00	\$674	\$977.5 bn	1450.9 m
4	Russia	Eastern Europe	0.11	\$3,722	\$534.3 bn	143.5 m
5	Mexico	Central America	0.22	\$3,310	\$433.2 bn	130.9 m
6	Japan	Eastern Asia	0.67	\$3,249	\$402.8 bn	124.0 m
7	South Korea	Eastern Asia	0.33	\$7,248	\$375.1 bn	51.8 m
8	United Kingdom	Northern Europe	0.67	\$5,325	\$368.6 bn	69.2 m
9	Brazil	South America	0.44	\$1,718	\$364.3 bn	212.0 m
10	Türkiye	Western Asia	0.11	\$3,973	\$339.8 bn	85.5 m
11	Saudi Arabia	Western Asia	0.00	\$8,780	\$310.0 bn	35.3 m
12	Indonesia	South-eastern Asia	0.33	\$985	\$279.3 bn	283.5 m
13	Italy	Southern Europe	0.67	\$4,039	\$238.1 bn	59.0 m
14	France	Western Europe	0.78	\$3,074	\$210.7 bn	68.6 m
15	Germany	Western Europe	0.89	\$1,870	\$156.2 bn	83.5 m
16	Canada	Northern America	0.78	\$3,623	\$149.6 bn	41.3 m
17	United Arab Emirates	Western Asia	0.00	\$12,568	\$138.1 bn	11.0 m
18	Thailand	South-eastern Asia	0.11	\$1,837	\$131.6 bn	71.7 m
19	Vietnam	South-eastern Asia	0.00	\$1,179	\$119.1 bn	101.0 m
20	Iran	Southern Asia	0.00	\$1,298	\$118.8 bn	91.6 m
21	Australia	Australia and New	0.78	\$4,307	\$117.1 bn	27.2 m
22	Spain	Southern Europe	0.78	\$2,355	\$115.0 bn	48.8 m
23	Bangladesh	Southern Asia	0.00	\$648	\$112.5 bn	173.6 m
24	Israel	Western Asia	0.33	\$10,835	\$108.1 bn	10.0 m
25	Argentina	South America	0.44	\$2,328	\$106.4 bn	45.7 m
26	Pakistan	Southern Asia	0.00	\$370	\$92.9 bn	251.3 m
27	Poland	Eastern Europe	0.67	\$2,510	\$91.8 bn	36.6 m
28	Egypt	Northern Africa	0.22	\$779	\$90.8 bn	116.5 m
29	Philippines	South-eastern Asia	0.44	\$664	\$76.9 bn	115.8 m
30	Kazakhstan	Central Asia	0.00	\$3,539	\$72.9 bn	20.6 m
31	Iraq	Western Asia	0.00	\$1,518	\$69.9 bn	46.0 m
32	Colombia	South America	0.44	\$1,320	\$69.8 bn	52.9 m
33	South Africa	Southern Africa	0.44	\$1,045	\$66.9 bn	64.0 m
34	Algeria	Northern Africa	0.22	\$1,342	\$62.8 bn	46.8 m
35	Peru	South America	0.33	\$1,690	\$57.8 bn	34.2 m
36	Malaysia	South-eastern Asia	0.56	\$1,583	\$56.3 bn	35.6 m
37	Singapore	South-eastern Asia	0.67	\$9,067	\$54.7 bn	6.0 m
38	Qatar	Western Asia	0.22	\$17,894	\$51.1 bn	2.9 m

#	Country	Region	R	Per Capita Loss	Total Loss (USD)	Population
39	Nigeria	Western Africa	0.33	\$217	\$50.5 bn	232.7 m
40	Ukraine	Eastern Europe	0.22	\$1,258	\$47.6 bn	37.9 m
41	Netherlands	Western Europe	0.89	\$2,251	\$40.5 bn	18.0 m
42	Romania	Eastern Europe	0.67	\$2,008	\$38.3 bn	19.1 m
43	Austria	Western Europe	0.78	\$3,885	\$35.7 bn	9.2 m
44	Ethiopia	Eastern Africa	0.22	\$265	\$34.9 bn	132.1 m
45	Kuwait	Western Asia	0.33	\$6,544	\$32.0 bn	4.9 m
46	Switzerland	Western Europe	0.89	\$3,467	\$31.2 bn	9.0 m
47	Venezuela	South America	0.00	\$1,054	\$30.0 bn	28.4 m
48	Uzbekistan	Central Asia	0.11	\$790	\$28.7 bn	36.4 m
49	Guatemala	Central America	0.11	\$1,538	\$28.3 bn	18.4 m
50	Morocco	Northern Africa	0.44	\$692	\$26.4 bn	38.1 m
51	Greece	Southern Europe	0.67	\$2,463	\$25.6 bn	10.4 m
52	Oman	Western Asia	0.22	\$4,733	\$25.0 bn	5.3 m
53	Ecuador	South America	0.33	\$1,375	\$24.9 bn	18.1 m
54	Kenya	Eastern Africa	0.33	\$426	\$24.1 bn	56.4 m
55	Hungary	Eastern Europe	0.67	\$2,329	\$22.3 bn	9.6 m
56	Dominican Republic	Caribbean	0.44	\$1,813	\$20.7 bn	11.4 m
57	Angola	Middle Africa	0.33	\$533	\$20.2 bn	37.9 m
58	Sri Lanka	Southern Asia	0.33	\$903	\$19.8 bn	21.9 m
59	Belarus	Eastern Europe	0.00	\$2,079	\$19.0 bn	9.1 m
60	Bulgaria	Eastern Europe	0.44	\$2,933	\$18.9 bn	6.4 m
61	Azerbaijan	Western Asia	0.00	\$1,821	\$18.6 bn	10.2 m
62	Myanmar	South-eastern Asia	0.00	\$340	\$18.5 bn	54.5 m
63	DR Congo	Middle Africa	0.11	\$162	\$17.7 bn	109.3 m
64	Côte d'Ivoire	Western Africa	0.33	\$546	\$17.4 bn	31.9 m
65	Serbia	Southern Europe	0.44	\$2,280	\$15.0 bn	6.6 m
66	Slovakia	Eastern Europe	0.67	\$2,599	\$14.1 bn	5.4 m
67	Bolivia	South America	0.11	\$1,105	\$13.7 bn	12.4 m
68	Cameroon	Middle Africa	0.11	\$458	\$13.3 bn	29.1 m
69	Turkmenistan	Central Asia	0.00	\$1,714	\$12.8 bn	7.5 m
70	Uganda	Eastern Africa	0.22	\$252	\$12.6 bn	50.0 m
71	Sudan	Northern Africa	0.00	\$246	\$12.4 bn	50.4 m
72	Libya	Northern Africa	0.00	\$1,642	\$12.1 bn	7.4 m
73	Bahrain	Western Asia	0.00	\$7,413	\$11.8 bn	1.6 m
74	Cambodia	South-eastern Asia	0.00	\$657	\$11.6 bn	17.6 m
75	Czechia	Eastern Europe	0.89	\$1,061	\$11.6 bn	10.9 m
76	Ghana	Western Africa	0.56	\$319	\$11.0 bn	34.4 m

#	Country	Region	R	Per Capita Loss	Total Loss (USD)	Population
77	Tanzania	Eastern Africa	0.56	\$158	\$10.8 bn	68.6 m
78	Jordan	Western Asia	0.33	\$924	\$10.7 bn	11.6 m
79	Portugal	Southern Europe	0.89	\$976	\$10.4 bn	10.7 m
80	Zimbabwe	Eastern Africa	0.22	\$583	\$9.7 bn	16.6 m
81	Honduras	Central America	0.11	\$857	\$9.3 bn	10.8 m
82	Paraguay	South America	0.33	\$1,283	\$8.9 bn	6.9 m
83	El Salvador	Central America	0.00	\$1,395	\$8.8 bn	6.3 m
84	New Zealand	Australia and New	0.89	\$1,640	\$8.7 bn	5.3 m
85	Panama	Central America	0.67	\$1,916	\$8.7 bn	4.5 m
86	Nepal	Southern Asia	0.33	\$289	\$8.6 bn	29.7 m
87	Tunisia	Northern Africa	0.56	\$557	\$6.8 bn	12.3 m
88	Papua New Guinea	Melanesia	0.33	\$601	\$6.4 bn	10.6 m
89	Haiti	Caribbean	0.11	\$536	\$6.3 bn	11.8 m
90	Guinea	Western Africa	0.00	\$424	\$6.3 bn	14.8 m
91	Croatia	Southern Europe	0.78	\$1,603	\$6.2 bn	3.9 m
92	Georgia	Western Asia	0.44	\$1,540	\$5.7 bn	3.7 m
93	Mozambique	Eastern Africa	0.00	\$164	\$5.7 bn	34.6 m
94	Lithuania	Northern Europe	0.78	\$1,959	\$5.7 bn	2.9 m
95	Senegal	Western Africa	0.44	\$296	\$5.5 bn	18.5 m
96	Burkina Faso	Western Africa	0.22	\$229	\$5.4 bn	23.5 m
97	Mali	Western Africa	0.33	\$219	\$5.4 bn	24.5 m
98	Trinidad and Tobago	Caribbean	0.33	\$3,747	\$5.1 bn	1.4 m
99	Zambia	Eastern Africa	0.33	\$237	\$5.1 bn	21.3 m
100	Nicaragua	Central America	0.00	\$712	\$4.9 bn	6.9 m
101	Chad	Middle Africa	0.00	\$240	\$4.9 bn	20.3 m
102	Slovenia	Southern Europe	0.78	\$2,287	\$4.9 bn	2.1 m
103	Afghanistan	Southern Asia	0.00	\$103	\$4.4 bn	42.6 m
104	Jamaica	Caribbean	0.33	\$1,551	\$4.4 bn	2.8 m
105	Madagascar	Eastern Africa	0.11	\$136	\$4.4 bn	32.0 m
106	Armenia	Western Asia	0.44	\$1,426	\$4.3 bn	3.0 m
107	Laos	South-eastern Asia	0.00	\$531	\$4.1 bn	7.8 m
108	Lebanon	Western Asia	0.33	\$696	\$4.0 bn	5.8 m
109	Niger	Western Africa	0.33	\$147	\$4.0 bn	27.0 m
110	Bahamas	Caribbean	0.00	\$9,864	\$4.0 bn	0.4 m
111	Bosnia and Herzegovina	Southern Europe	0.56	\$1,248	\$3.9 bn	3.2 m
112	Congo Republic	Middle Africa	0.00	\$621	\$3.9 bn	6.3 m
113	Botswana	Southern Africa	0.33	\$1,539	\$3.9 bn	2.5 m
114	Brunei Darussalam	South-eastern Asia	0.00	\$8,288	\$3.8 bn	0.5 m

#	Country	Region	R	Per Capita Loss	Total Loss (USD)	Population
115	Albania	Southern Europe	0.56	\$1,517	\$3.6 bn	2.4 m
116	Rwanda	Eastern Africa	0.00	\$250	\$3.6 bn	14.3 m
117	Tajikistan	Central Asia	0.00	\$335	\$3.6 bn	10.6 m
118	Cyprus	Western Asia	0.78	\$2,578	\$3.5 bn	1.4 m
119	Kyrgyzstan	Central Asia	0.33	\$484	\$3.5 bn	7.2 m
120	Mauritius	Eastern Africa	0.22	\$2,798	\$3.5 bn	1.2 m
121	Gabon	Middle Africa	0.44	\$1,372	\$3.5 bn	2.5 m
122	North Macedonia	Southern Europe	0.33	\$1,858	\$3.4 bn	1.8 m
123	Guyana	South America	0.56	\$3,957	\$3.3 bn	0.8 m
124	Equatorial Guinea	Middle Africa	0.00	\$1,686	\$3.2 bn	1.9 m
125	Costa Rica	Central America	0.89	\$620	\$3.2 bn	5.1 m
126	Mongolia	Eastern Asia	0.56	\$900	\$3.2 bn	3.5 m
127	Somalia	Eastern Africa	0.11	\$157	\$3.0 bn	19.0 m
128	Benin	Western Africa	0.56	\$198	\$2.9 bn	14.5 m
129	Monaco	Western Europe	0.00	\$72,000	\$2.8 bn	39 k
130	Mauritania	Western Africa	0.11	\$528	\$2.7 bn	5.2 m
131	Togo	Western Africa	0.00	\$280	\$2.7 bn	9.5 m
132	Malawi	Eastern Africa	0.33	\$105	\$2.3 bn	21.7 m
133	Liechtenstein	Western Europe	0.00	\$51,695	\$2.1 bn	40 k
134	Namibia	Southern Africa	0.56	\$588	\$1.8 bn	3.0 m
135	Malta	Southern Europe	0.78	\$2,927	\$1.7 bn	0.6 m
136	Maldives	Southern Asia	0.33	\$2,676	\$1.4 bn	0.5 m
137	Sierra Leone	Western Africa	0.33	\$161	\$1.4 bn	8.6 m
138	Moldova	Eastern Europe	0.78	\$505	\$1.2 bn	2.4 m
139	Eswatini	Southern Africa	0.22	\$912	\$1.1 bn	1.2 m
140	Iceland	Northern Europe	0.89	\$2,868	\$1.1 bn	0.4 m
141	Andorra	Southern Europe	0.00	\$12,326	\$1.0 bn	82 k
142	Djibouti	Eastern Africa	0.22	\$829	\$969 m	1.2 m
143	Liberia	Western Africa	0.33	\$170	\$956 m	5.6 m
144	Montenegro	Southern Europe	0.67	\$1,326	\$827 m	0.6 m
145	Belize	Central America	0.00	\$1,920	\$801 m	0.4 m
146	Fiji	Melanesia	0.56	\$857	\$796 m	0.9 m
147	Burundi	Eastern Africa	0.00	\$55	\$771 m	14.0 m
148	Central African Republic	Middle Africa	0.11	\$129	\$688 m	5.3 m
149	St. Lucia	Caribbean	0.00	\$3,545	\$637 m	0.2 m
150	Bhutan	Southern Asia	0.33	\$766	\$607 m	0.8 m
151	Suriname	South America	0.56	\$928	\$589 m	0.6 m
152	Guinea-Bissau	Western Africa	0.11	\$252	\$555 m	2.2 m

#	Country	Region	R	Per Capita Loss	Total Loss (USD)	Population
153	Antigua and Barbuda	Caribbean	0.00	\$5,886	\$552 m	94 k
154	San Marino	Southern Europe	0.00	\$14,970	\$509 m	34 k
155	Barbados	Caribbean	0.78	\$1,770	\$500 m	0.3 m
156	Lesotho	Southern Africa	0.33	\$194	\$454 m	2.3 m
157	Comoros	Eastern Africa	0.11	\$416	\$360 m	0.9 m
158	Grenada	Caribbean	0.00	\$2,926	\$343 m	0.1 m
159	Gambia	Western Africa	0.56	\$116	\$321 m	2.8 m
160	Solomon Islands	Melanesia	0.33	\$387	\$317 m	0.8 m
161	Samoa	Polynesia	0.00	\$1,348	\$294 m	0.2 m
162	St. Vincent and the Grenadin	Caribbean	0.00	\$2,875	\$289 m	0.1 m
163	St. Kitts and Nevis	Caribbean	0.00	\$5,990	\$281 m	47 k
164	Cabo Verde	Western Africa	0.67	\$519	\$273 m	0.5 m
165	Timor-Leste	South-eastern Asia	0.67	\$133	\$187 m	1.4 m
166	Dominica	Caribbean	0.00	\$2,601	\$172 m	66 k
167	Vanuatu	Melanesia	0.56	\$455	\$149 m	0.3 m
168	Tonga	Polynesia	0.00	\$1,413	\$147 m	0.1 m
169	Sao Tome and Principe	Middle Africa	0.44	\$582	\$137 m	0.2 m
170	Micronesia, Fed. Sts.	Micronesia	0.00	\$1,042	\$118 m	0.1 m
171	Kiribati	Micronesia	0.11	\$572	\$77 m	0.1 m
172	Marshall Islands	Micronesia	0.00	\$1,932	\$73 m	38 k
173	Seychelles	Eastern Africa	0.89	\$595	\$72 m	0.1 m
174	Palau	Micronesia	0.00	\$3,903	\$69 m	18 k
175	Nauru	Polynesia	0.00	\$3,402	\$41 m	12 k
176	Tuvalu	Polynesia	0.00	\$1,586	\$15 m	10 k